

Stakeholder Engagement Plan

Akagera Group

Project: Development of a Small Language Model for Rural Farmers in Nigeria

1. Stakeholder Profile: Rural Farmers

Stakeholder: Rural smallholder farmers across key agricultural regions of Nigeria, including women-led farming households and farmer cooperatives.

Power / Interest: High interest, currently low formal power in the design process. This plan is intended to shift farmers toward a co-design role rather than a passive end-user role.

Interest / value: Farmers want practical support in navigating climate change, pest and soil challenges, and evolving agricultural practices in order to improve yield and income. Their core interest is in tools that strengthen and extend local farming knowledge and the guidance of trusted human extension workers, not tools that sideline or replace that expertise. They value plain-language, locally relevant advice delivered in a form they can access with limited connectivity or digital literacy.

Potential harms: The model may misinterpret local practices, offer advice that is agronomically inappropriate for a specific microclimate or soil type, or gradually diminish the role and authority of human extension workers. Overreliance on the tool could also widen gaps for farmers with low digital literacy, limited connectivity, or minority-language backgrounds who are less able to use or correct the system.

Potential benefits: Farmers gain a tailored, on-demand knowledge tool that complements extension workers, improves timely access to climate and agronomic guidance, and helps them respond faster to emerging risks (e.g., pest outbreaks, weather shifts). Over time this can improve yields, reduce input waste, and strengthen farmers' agency in adopting new practices on their own terms.

Engagement method: Participatory sensing, paired with in-person workshops and a standing farmer advisory committee. Participatory sensing enables the collection of representative local data to train and validate the model, while the advisory committee and workshops ensure farmers actively shape the tool's scope, tone, and safeguards rather than only supplying data.

Plan use: Farmer participation ensures the model reflects local linguistic and cultural nuance, builds trust and legitimacy within communities, and surfaces harms early enough to correct them. Findings from each engagement cycle will directly inform model revisions, extension-worker training materials, and the project's go/no-go decisions at each phase gate.

2. Engagement Timeline

A phased, 12-month engagement approach ensures farmers are involved from problem definition through to scale-up, rather than only at data collection.

Phase	Key Activities	Milestone	Output
Phase 1: Groundwork (Months 1-2)	Conduct baseline surveys and focus group discussions with farmer	Community entry, local advisory committee	Farmer needs assessment report; list of engaged cooperatives

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	cooperatives; map local languages, dialects, and dominant farming practices across target regions.	formed, informed-consent protocol agreed	
Phase 2: Co-design (Months 3-4)	Run participatory sensing workshops where farmers help define the questions, terms, and scenarios the model should handle; pilot early prototypes with a small user group.	First data collection round completed; farmers validate initial model responses	Curated local-language dataset; farmer feedback log
Phase 3: Field Testing (Months 5-7)	Deploy the model with extension workers and volunteer farmers via SMS/USSD or voice channels; collect usage data and satisfaction feedback.	Field trial across 3-5 communities; extension workers trained as co-facilitators	Field trial report; revised model version
Phase 4: Feedback & Iteration (Months 8-9)	Hold community feedback sessions to review harms/benefits observed; adjust model outputs, tone, and coverage based on findings.	Grievance and feedback mechanism operational; corrective actions logged	Updated engagement log; model refinement notes
Phase 5: Scale & Sustain (Months 10-12)	Expand to additional communities; formalize ongoing engagement (e.g., quarterly community advisory sessions) and handover plan with local partners.	Scale-up plan endorsed by advisory committee; sustainability plan agreed	Scale-up roadmap; sustainability and handover plan

3. Risk Assessment

Risks specific to the rural farmer stakeholder group, rated by likelihood and impact, with corresponding mitigations.

Risk	Likelihood	Impact	Mitigation
AI misinterprets local agricultural practices or terminology	Medium	High	Co-design content with local agronomists and farmer representatives; validate model outputs through community review cycles before wide release.

Risk	Likelihood	Impact	Mitigation
Erosion of the human extension worker's role and trusted relationships	Medium	High	Position the tool as a supplement, not a replacement, for extension workers; involve extension workers as co-facilitators and trainers rather than bypassing them.
Low digital literacy or connectivity limits access and equitable participation	High	Medium	Use low-bandwidth channels (SMS/USSD/voice); pair digital access with in-person community sessions; provide device/data support where feasible.
Language or dialect gaps reduce model accuracy for some sub-groups	Medium	Medium	Recruit a diverse pool of language contributors during data collection; track accuracy by dialect and prioritize underserved groups in iteration cycles.
Data privacy concerns around farm-level or personal data	Low	High	Apply data minimization and informed consent; anonymize datasets; communicate data use plainly in the farmers' own language.
Engagement fatigue or tokenistic participation	Medium	Medium	Ensure two-way feedback loops with visible follow-through; compensate farmers' time where appropriate; report back on how their input shaped the tool.

4. Success Metrics (KPIs)

Metrics to track whether engagement with rural farmers is genuine, effective, and translating into a safer, more useful model.

KPI	Target	Measurement Method
Farmer participation rate	≥ 70% of invited farmers/cooperatives actively engaged per phase	Attendance and participation logs from workshops and sensing sessions
Model accuracy on local terminology/dialects	≥ 85% farmer-rated accuracy by end of field testing	Farmer validation surveys; sample review by local agronomists
Trust and perceived value	≥ 75% of surveyed farmers report the tool is useful and trustworthy	Pre/post engagement surveys; qualitative interviews

KPI	Target	Measurement Method
Extension worker collaboration	100% of trial communities have an extension worker co-facilitating	Field trial staffing records
Grievance/feedback resolution	$\geq 90\%$ of raised concerns acknowledged within 2 weeks	Feedback mechanism tracker
Harm incidents identified and mitigated	0 unresolved harm reports at project close	Incident log and mitigation tracker